Burden and Trends in Sexually Transmitted Infections Across the European Union (EU-27) and the United Kingdom From 2000-2021

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Introduction

Sexually transmitted infections (STIs) are caused by pathogens whose primary transmission route is through sexual contact. STIs represent a significant global public health issue, with the World Health Organization estimating that more than one million new infections occur every day worldwide[1]. The World Health Organization also estimates that there were 374 million new infections of chlamydia, gonorrhoea, syphilis, or trichomoniasis in people 15-49 years of age in 2020 across the world[1]. These infections have a large variety of symptoms and can result in serious consequences, such as infertility, cancers, and pregnancy complications, on the fertility and sexual health of the affected individual[1]. The risk of perinatal transmission is higher in STIs with asymptomatic infections, such as chlamydia where up to 70% of infections remain undetected and gonorrhoea where up to 50% of infections remain undetected[2]. Prevention efforts, such as pre-exposure prophylaxis for human immunodeficiency virus (HIV) and doxycycline post-exposure prophylaxis for chlamydia and syphilis, have attempted to quell the spread of STIs[3,4]. Other efforts have included sexual education, the development of STI surveillance systems, presenting sexuality as a “positive human experience” to reduce stigma, and improved access to testing and treatment[3]. However, these efforts have been hindered by health disparities, limited resources, data gaps, and lack of robust surveillance systems. In addition, there has been limited uptake of some prophylaxis treatments due to drug-related costs, lack of national guidelines, hesitancy of healthcare providers to offer prophylaxis treatments, lack of awareness of treatment availability, and self-perceived risks[3,4].

The 27 member states of the European Union (EU-27) and the United Kingdom (UK), collectively called the EU-27+UK, are not immune to the hindrance of STI prevention efforts. A 2024 European Centre for Disease Prevention and Control (ECDC) report estimates that approximately 300,000 new cases of bacterial STIs are reported annually to the European Surveillance System[5]. However, the report states that this estimate may be incorrect due to variations in surveillance system coverage, screening policies, and testing practices[5]. A 2019 ECDC report stated that there were over 400,000 new cases of chlamydia alone throughout the EU-27+UK in 2017[2]. Reported incidence, prevalence, and death rates vary across other local, national, and global organisations due to differences in STI surveillance and reporting[6–8].

In addition, some countries do not require notification for certain STIs. Spain does not require notification for herpes simplex virus despite an estimated 10-15% of the adult population serving as carriers[6]. The lack of a requirement for notification has led to an underestimation for the actual incidence of the disease[6]. Underestimation continues to be an issue due to under diagnosis, particularly in areas with limited resources. A sharp decline in the number of new cases of STIs was noted for the years 2020-2021 due to the COVID-19 pandemic[7]. One study in Catalonia found that there were approximately 51% less STI cases reported than expected during the first five months of the lockdown[7]. The sharp decline was attributed to under diagnosis and under reporting as a rapid increase in the number of new infections has been noted throughout the EU-27+UK since 2021[5–7].

The aim of this retrospective, descriptive study was to assess the STI disease burden, estimated in deaths, disability-adjusted life years (DALYs), prevalence, and incidence, in the EU-27+UK and to compare STIs between these countries and over time from 2000 to 2021 using secondary data from the Global Burden of Disease (GBD) Study 2021.

Materials and Methods

Data Source

Secondary data was obtained from the Institute for Health Metrics and Evaluation's Global Burden of Disease (GBD) Study 2021 data resources using the GBD results tool. The GBD study gathers health data from multiple sources, such as hospitals, governments, and various other health databases across the world[9]. That data is then synthesised into a single repository that allows for the production of estimates, data visuals, and publications[9]. The GBD results tool allows users to obtain data from the repository based on selected filters.

For this study, we selected "cause of death or injury" for the source of estimates. The selected measures were: Deaths, DALYs, prevalence, incidence. The selected metrics were: number, percent, rate. The selected causes were: HIV/AIDS, syphilis, chlamydial infection, gonococcal infection, trichomoniasis, genital herpes, other sexually transmitted infections. Locations selected included all of the EU-27 member countries and the United Kingdom. Selected age ranges included: 0-14 years, 15-49 years, 50-74 years, 75+ years, "All Ages." Selected sexes included: male, female, both. Years selected included all years from 2000 to 2021. Data files with data for the applied filters were downloaded from the GBD repository in comma separated value (CSV) format. The amount of data per download was limited based on GBD restrictions, so data was downloaded using the selected filters for measures, metrics, causes, age ranges, and sexes for each individual country in increments of 5-6 years. Downloaded CSV files were saved into a single folder.

Data Cleaning, Analysis, and Visualization

Data analysis was completed using the R programming language within the RStudio 2024.09.0+375 "Cranberry Hibiscus" Release environment. Each of the individual CSV files was loaded into RStudio and then combined into a single CSV file while ensuring data integrity. The combined CSV file was then loaded into RStudio and descriptive statistics reports and summaries were utilised to determine potential outliers, missing values, and unreliable data.

Outliers were determined using a combination of functions to diagnose the quality of data as well as box plots with the interquartile range and the 25th and 75th percentiles. Missing values were identified using conditional counting functions. Unreliable data was determined by confirming the metric measurements as defined in the GBD guide (i.e., rates described as “events per 100,000 population”) and analysing nonsensical data values (such as prevalence rates == “0”).

Following the initial cleaning of the data, we then began a full analysis of the data starting with plotting several graphs for each measure and metric for all causes across all countries and all years to get an overview and identify any areas of potential interest. Bar charts, line graphs, and choropleth maps were used to visualise trends. Any identified areas of potential interest were then analysed more closely. The overall results were then compared against World Health Organization and ECDC reports.

Results

Data Quality Analysis

With the parameters stated above we obtained 702240 observations from the GBD results tool.

We proceeded to summarise the data and analyse it for missing data, outliers, and unreliable data. The Missing Data analysis returned no missing data in the database. The initial outlier analysis returned over 143394 outliers, which we found to be too much for a database of 702240 values. We charted a boxplot with all outliers which showed that the outliers were concentrated in specific categories, meaning we were not filtering the correct values to test for outliers. Once we applied the correct filters, all tests found no outliers.

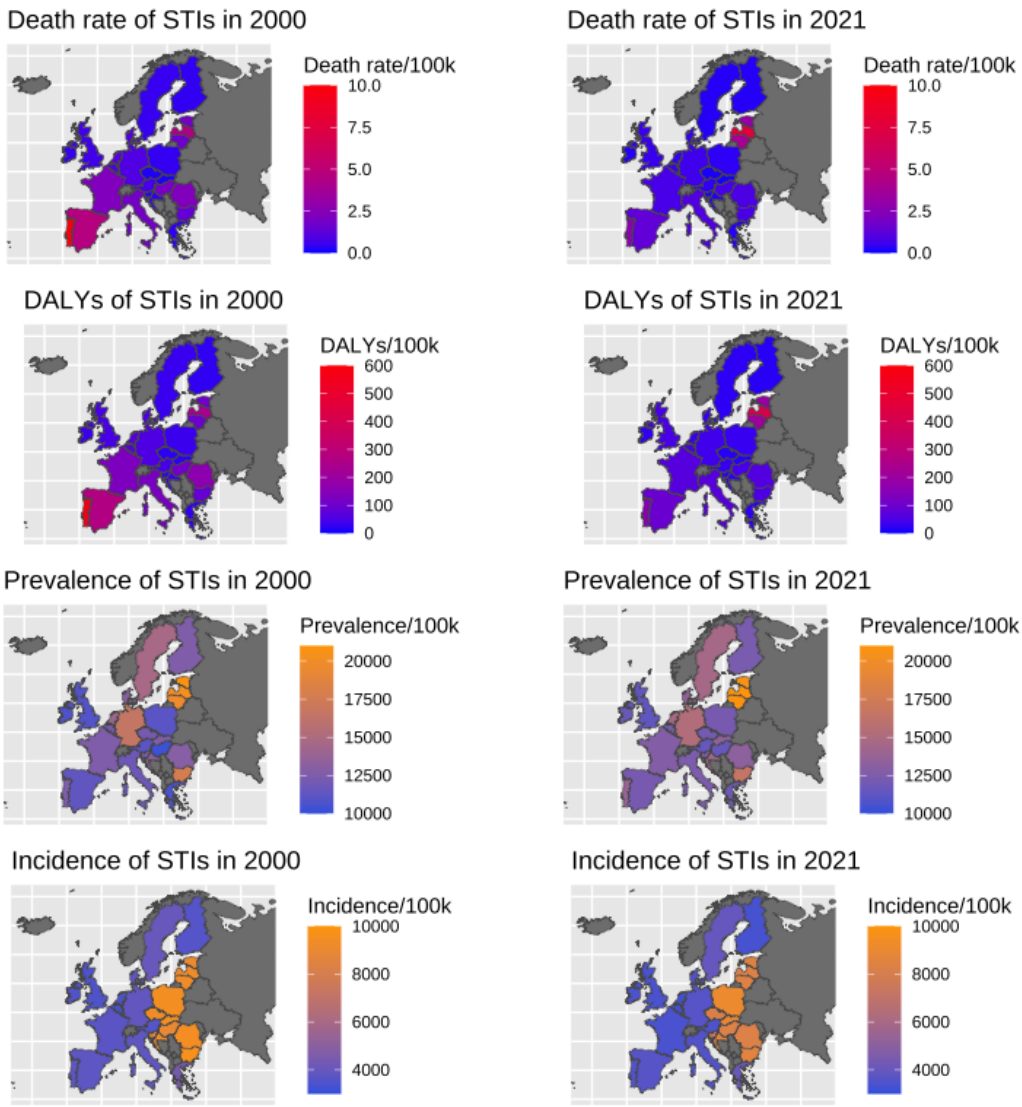
Regarding unreliable data, we checked for nonsensical values (such as rates per 100000 people greater than 100000, percentages >100, negative values, values = 0) and found that there were 36960 “zero” values in the data set (approximately 5.26% of total data points). Upon closer analysis, discovered that the database lacked information on: - Deaths caused by Chlamydial infection on Males - Incidence of Syphilis on ages 75+ - Incidence of “Other sexually transmitted infections” - Prevalence of “Other sexually transmitted infections” on ages 75+ We interpreted these zero values as unreported data, but nonetheless decided to keep them in the database, as they were a relatively minor subset of values, and their available mirrored values showed that they had an overall negligible impact on the analysis. No other unreliable data was detected.

Data Analysis

Choropleth Maps

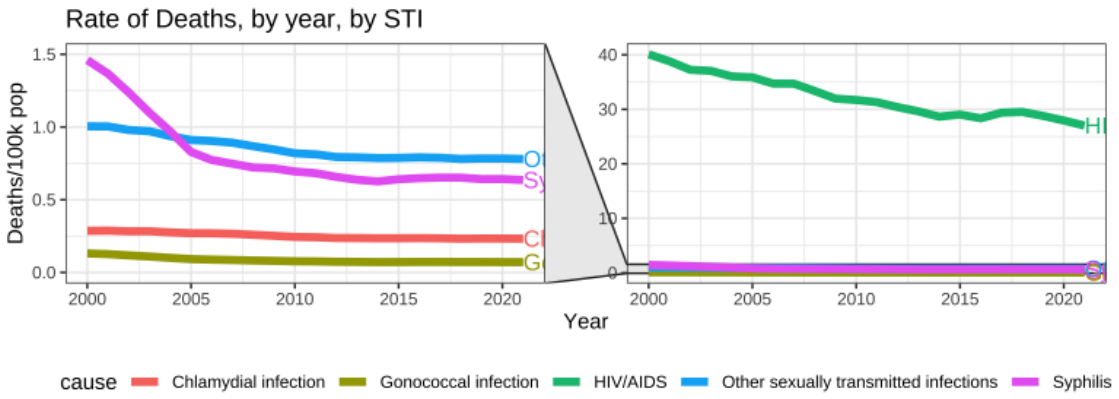
We built choropleth maps using the Deaths/DALYs/Prevalence/Incidence Rates at the ends of the time periods in analysis to show the evolution of STIs across 21 years. The maps showed that overall Death and DALY Rate decreased across the EU-27+UK with the exception of the Baltic countries.

Prevalence and Incidence of STIs across the EU-27+UK have kept stable during the period in study. One notable finding is the sustained higher incidence of STIs in Eastern European countries (ie, former members of the Soviet block).



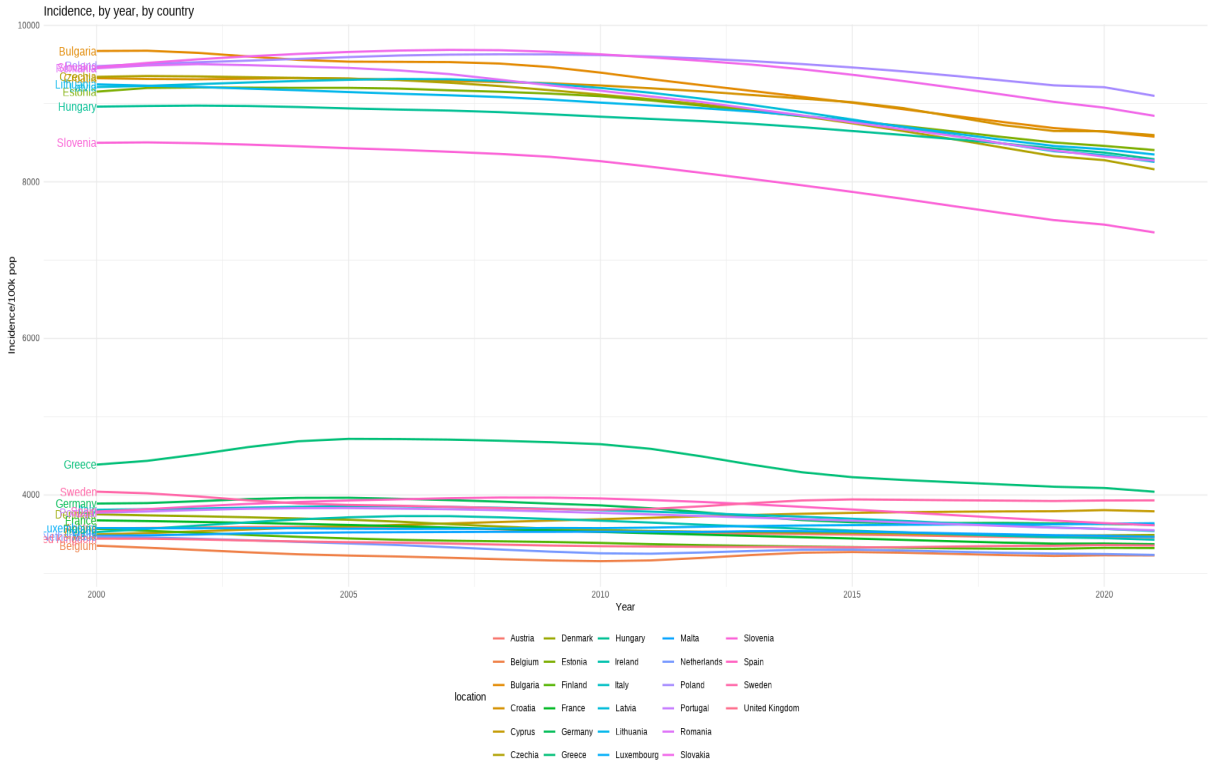
Detail of Death and DALY Rates

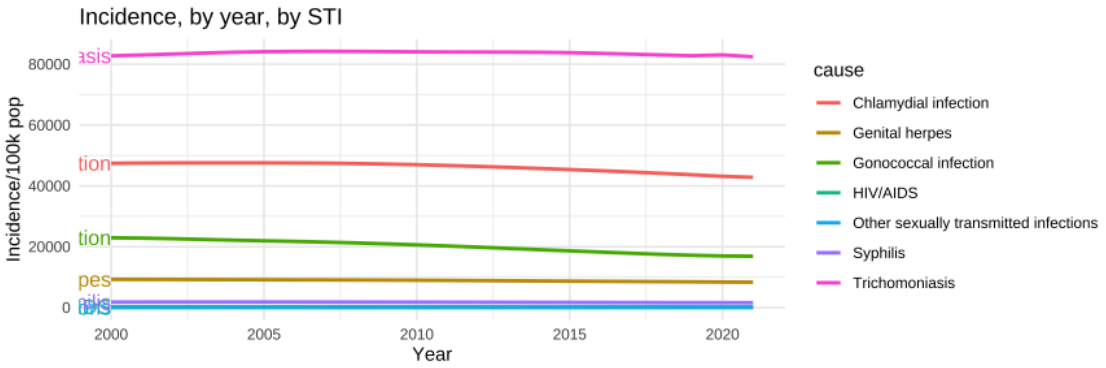
To better visualise these changes we generated graphs of overall death and DALY rates across time by STI and by country: these show a general decrease in STI deaths/DALYs across all countries with significant decreases in the burden caused by HIV/AIDS and Syphilis, which contrast with the steady increase in both STI deaths/DALYs rates in the Baltic countries.



Detail of Prevalence and Incidence Rates

Regarding prevalence and incidence, both rates have kept relatively stable throughout the years, with the overall incidence showing a slight decrease over time. These graphs allowed us to identify some interesting findings: first, Germany observed a very large variation of prevalence of Genital Herpes between 2010 and 2015 (which was not mirrored in its incidence); second, the incidence of all STIs was been slowly decreasing, with Trichomoniasis having the highest incidence of all STIs analysed; third, the Eastern/Western separation seen in the choropleth maps above can be clearly seen in the graph plotting the incidence of STIs by country and year.





Analysis by Sex and Age

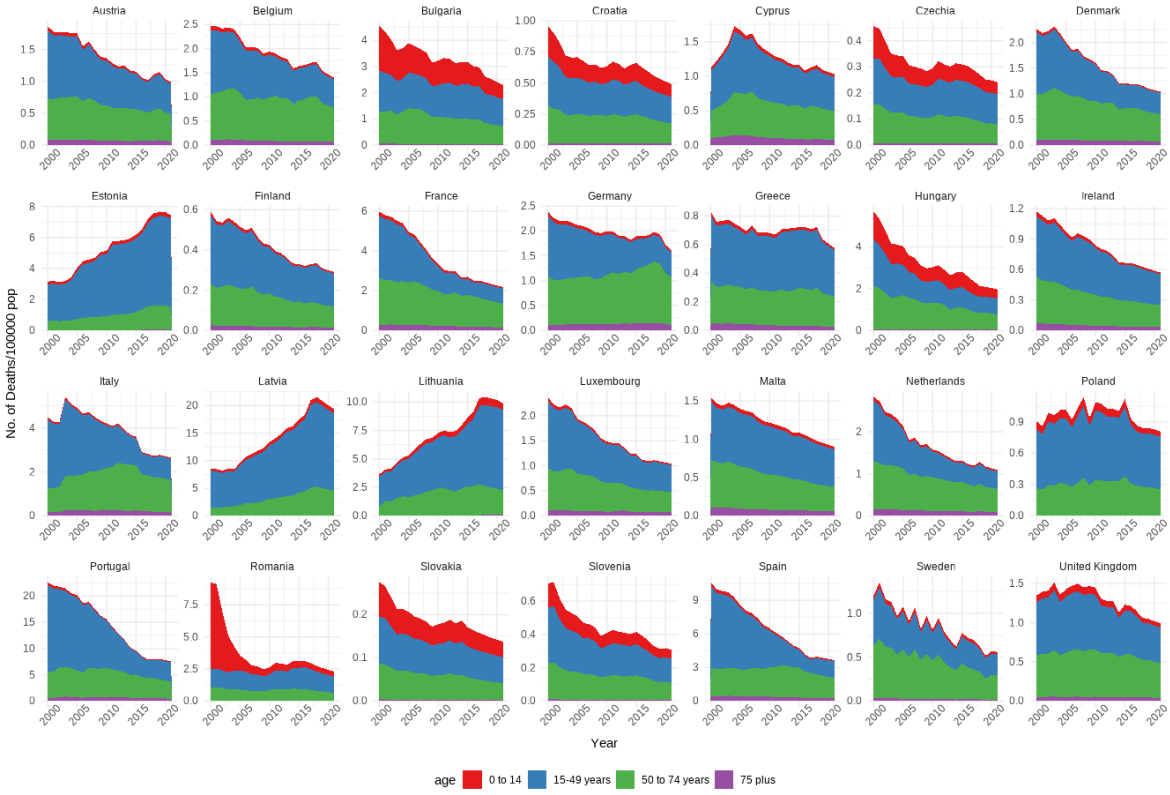
We built several faceted graphs to explore the age and sex distributions of STIs and found that they follow the overall trend pattern identified earlier.

Overall, Males are responsible for a higher proportion of deaths, DALYs, and Incidence regarding STIs, while in terms of prevalence the Males/Females ratio is more balanced (with a slight inclination towards Females). These proportions were kept relatively the same when analysing by individual STI and country.

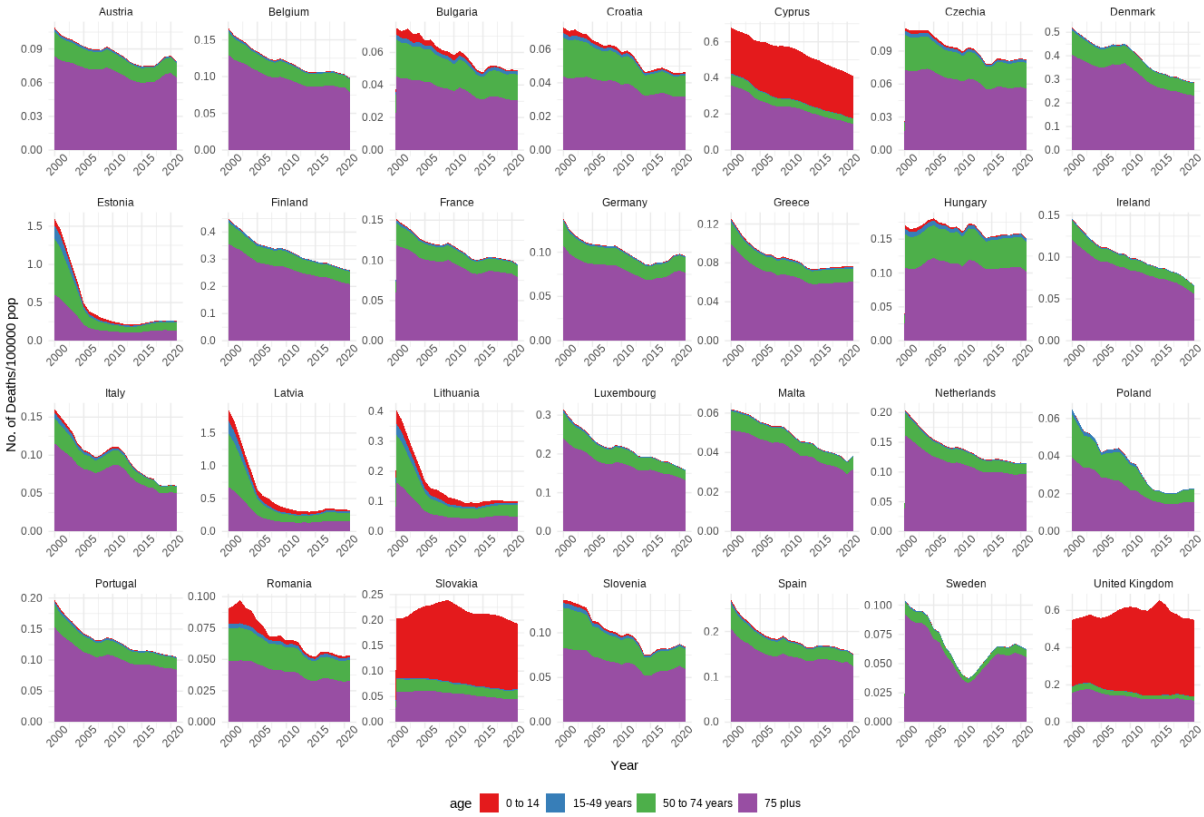
In overall analysis of age groups we noticed that the distribution in prevalence and incidence rates remained stable in absolute and relative terms. However the Death rates were the ones with the most differences between them: in general the death rate has fallen across the board, with most countries maintaining their relative distribution; notable exceptions are the Baltic countries whose rates have increased primarily due to the increase in the age group 15-49; Portugal, Spain, and France have seen their overall rates decrease significantly due to the high reduction in cases in the age group 15-49; and Romania displayed a similar situation to the previous group, but their group that most declined was the age group of 0-14.

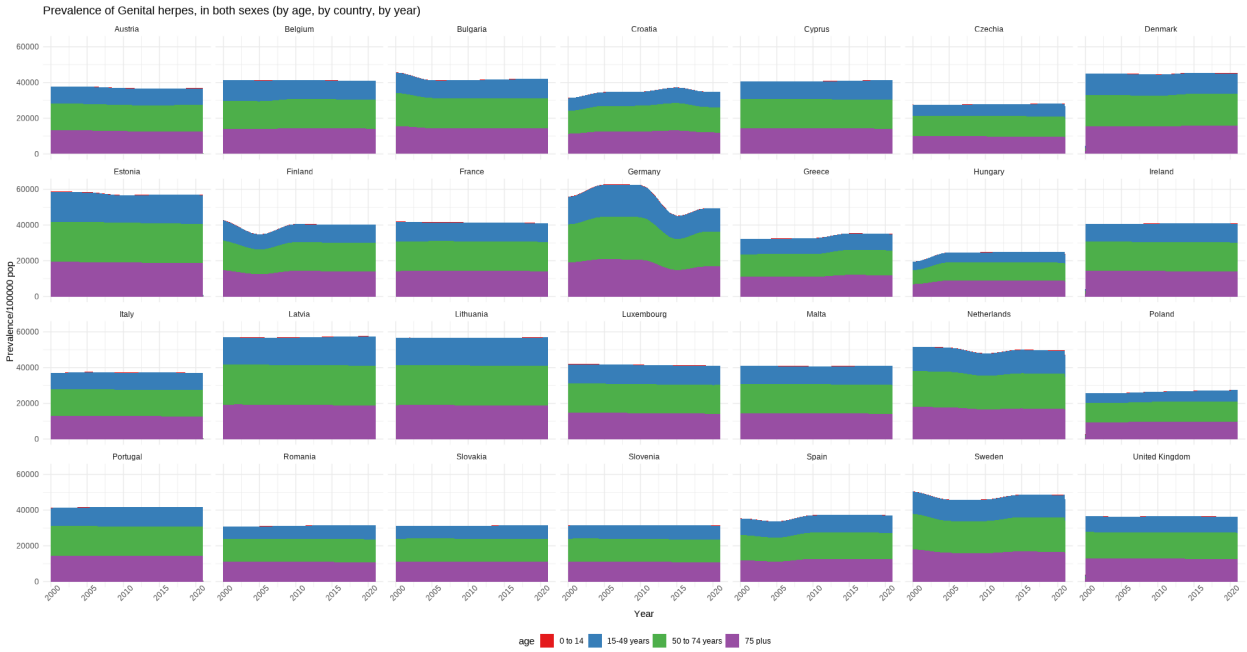
Each STI has its own pattern or Age distributions: HIV seems to mostly affect younger populations (ages 15-49 and 50-74) while Syphilis seems to mostly affect older populations (ages 50-74 and 75+). In all cases the relative proportion of age distribution was kept relatively similar throughout the years: a notable exception was Romania, which had a high death rate in the 2000s in younger patients, which was rapidly corrected until 2010 and kept the same proportions ever since.

Death Rate, caused by HIV/AIDS, in both sexes (by age, by country, by year)



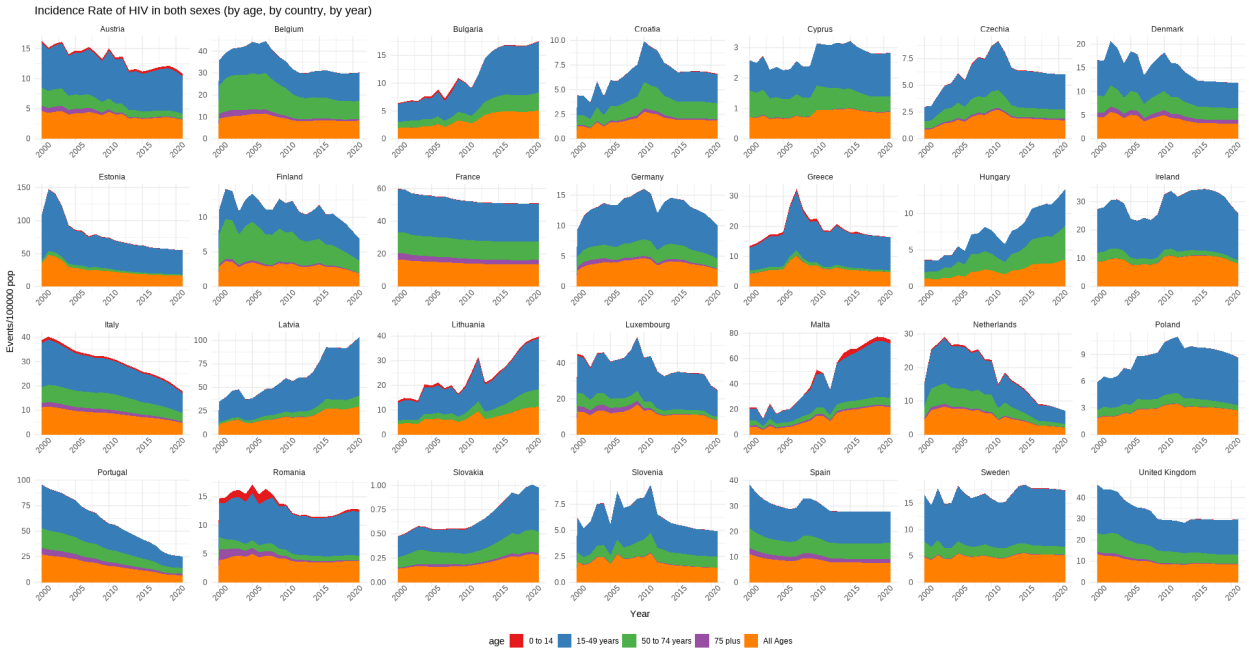
Death Rate, caused by Syphilis, in both sexes (by age, by country, by year)





Other Notable Findings

The faceted incidence plot of HIV/AIDS by age group (both sexes, by country, by year) shows interesting trends in several of the analysed countries: several countries registered unexpected spikes/decreases of HIV incidence for a couple of years, and returned to their usual values a couple of years later. There seems to be a correlation between these unexpected peaks and troughs and the Eurozone Crisis that affected the entire continent. There are however some notable exceptions in Portugal and Italy, which don't seem to have been affected by crisis effects on their healthcare systems.



Discussion

This work shows that deaths caused by STIs are decreasing from 2000 until 2021, indicating that the implemented health measures have had positive effects over time regarding STI-related deaths. However,there is heterogeneity across countries. Latvia still has high rates of deaths due to STIs, highlighting the need to reinforce measures against STIs in the country. Romania was another exception where high death rates were noted in the 2000s in younger patients, followed by a rapid decline until 2010 with death rates remaining stable since. In spite of showing a clear development since 2000, the overall incidence of STIs has remained stable, reinforcing the idea that better measures are needed to avoid transmission of STIs between individuals. Also, the incidence of STIs appears higher in Eastern European countries.

Additionally, prevalence also seems to be stable over time, pointing out that individuals are equally infected as before, but have less lethal consequences. The prevalence is homogeneously distributed in European countries, except for Slovenia, Latvia, and Lithuania, where clearly the prevalence is higher. A possible explanation is the absence of national guidelines related to diagnosis and management of STIs, as well as no orientations regarding the partner notification. In Latvia, for example, the diagnosis and management are delivered in consultations that are not free and must be co-payed by the individuals.

Another remarkable finding is that the rate of deaths caused by HIV/AIDs has dramatically decreased over the years, predominantly affecting those 15-49 years of age, which is in accordance with the data reported by the ECDC and World Health Organization. This is despite the incidence of HIV/AIDS remaining relatively stable with a slight upward trend in recent years, indicating that improved treatment and reducing the stigma associated with HIV has been effective at improving the prognosis for individuals with the disease. A potential explanation for a smaller proportion of individuals 50 years of age and over in our data is the former “lethality” associated with HIV/AIDS from the 1980s to early 2000s.

The slight upward trend in the incidence of HIV/AIDS across all countries could be attributed to the relaxation of anti-STI measures following the Eurozone Crisis that began in 2009. The relaxation of anti-STI measures could also explain the increasing prevalence of genital herpes in Germany from 2010-2015. Reduced testing, particularly in diseases such as herpes simplex virus that are seen as less “dangerous” to public health and do not demand active testing, following the Eurozone Crisis could be a factor for the increasing incidence and prevalence that we noted.

This work presents some limitations. The first limitation is related to the quality and availability of the data. The data available in GBD is pooled from multiple sources that each have different reporting methods, which is a likely explanation for the potentially missing data we noted earlier. In addition, data for the years 2020 and 2021 may not reflect the true numbers and rates as the COVID-19 pandemic likely caused under diagnosis and under reporting. The latest ECDC report from 2024 notes a sharp increase in the rates of new STIs since 2021. However, we were unable to assess this as we only had data available through 2021.

Additional studies with sufficient data are needed to validate these results.

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